

◆ Array (Special Object)

Stores multiple values in a single variable.

```
let fruits = ["Apple", "Banana", "Mango"];
```

🔗 Type Checking

```
typeof "Hello" // string
typeof 10      // number
typeof true   // boolean
```

🔗 Key Points

- Primitive → single value
- Non-Primitive → multiple/complex values
- JavaScript is **dynamically typed**
- Use `typeof` to check type

What is an Array in JavaScript?

An **array** is a special variable used to **store multiple values in a single variable**.

☞ Instead of creating many variables, we store all values together.

◆ Definition

An array is a **collection of elements stored in a continuous memory location**, accessed using an **index**.

◆ Example

```
let fruits = ["Apple", "Banana", "Mango"];
```

☞ Here:

- `fruits` → array name
 - `"Apple", "Banana", "Mango"` → elements
-

? How to Access Array Elements

Array elements are accessed using **index numbers**.

☞ Index starts from **0**

💡 Access Example

```
let fruits = ["Apple", "Banana", "Mango"];

console.log(fruits[0]); // Apple
console.log(fruits[1]); // Banana
console.log(fruits[2]); // Mango
```

💡 Modify Element

```
fruits[1] = "Orange";
```

☞ Now array becomes:

```
["Apple", "Orange", "Mango"]
```

💡 Loop Through Array

```
for(let i = 0; i < fruits.length; i++) {
  console.log(fruits[i]);
}
```

☞ Used to **access all elements easily**

? Key Points

- Array stores **multiple values in one variable**
- Index always starts from **0**
- Access using `array[index]`
- Useful for **looping and data handling**

5 . Difference between == and ===

- == compares **values only**, and performs **type conversion** if needed.
- === compares **both value and type** (strict comparison).

JavaScript Example

```
let a = 5;
let b = "5";

console.log(a == b);    // true   (value same, type ignored)
console.log(a === b);   // false  (type different: number vs string)
```

🔗 More Examples

```
console.log(0 == false);    // true
console.log(0 === false);   // false

console.log(null == undefined);    // true
console.log(null === undefined);   // false
```

🔗 Key Point (Easy Memory Trick)

🔗 == → **checks value only**

🔗 === → **checks value + type (strict)**

10 Use of the typeof operator

- The `typeof` operator is used to **find the data type** of a variable.
- Example:

```
let name = "Hari";
console.log(typeof name); // "string"
console.log(typeof 10);  // "number"
console.log(typeof true); // "boolean"
```

🔗 Helps in debugging and checking variable types.

10 Use of the typeof operator

- The `typeof` operator is used to **find the data type** of a variable.
- Example:

```
let name = "Hari";  
console.log(typeof name); // "string"  
console.log(typeof 10); // "number"  
console.log(typeof true); // "boolean"
```

☞ Helps in debugging and checking variable types.